Erratum: Adaptive False Discovery Rate Control with Privacy Guarantee

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1. Introduction

In Xia and Cai (2023), Xia and Cai introduce DP-AdaPT, which combines a novel p-value transformation preserving mirror-conservatism with a mirror-peeling algorithm that enables finite-sample FDR control. The correction in this note is limited to the privacy accounting of one implementation: the μ-GDP analysis of the Gaussian Report Noisy Min subroutine. It does not affect the novelty of the transformation, the mirror-peeling construction, the finite-sample FDR control result, or the validity of the (ε, δ)-DP mechanism.

The original paper presents two privacy formulations: a μ-GDP version and an (ε, δ)-DP version. This note identifies a gap only in the μ-GDP proof of the Report Noisy Min Algorithm. The separate (ε, δ)-DP mechanism is not affected. The purpose of this note is to provide a minimal correction. We replace the Gaussian noise used in the selection step with Laplace noise, as in the (ε, δ)-DP version. The Laplace Report Noisy Min mechanism satisfies pure differential privacy. The pure-DP guarantee is then converted to μ-GDP using the embedding of ε-DP into Gaussian differential privacy.

2. Assumptions, Algorithm, and Privacy Analysis

Throughout this section, let Φ denote the cumulative distribution function of the standard normal distribution. We write $\mathrm{Lap}(0, \lambda)$ for the Laplace distribution with mean 0 and scale parameter λ. We begin by recalling the relevant sensitivity definitions, then present the corrected algorithms.

Definition 1 (G-transformed sensitivity of p-values) *Let $p : \mathcal{S} \to (0, 1)$ be a deterministic p-value function. Its G-transformed sensitivity is defined as*

$$\Delta_G(p) := \sup_{\mathcal{S} \sim \mathcal{S}'} \left| G^{-1}\{p(\mathcal{S})\} - G^{-1}\{p(\mathcal{S}')\} \right|,$$

where $\mathcal{S}$ and $\mathcal{S}'$ are two neighboring data set. We say that p has G-transformed sensitivity at most Δ if $\Delta_G(p) \leq \Delta$.

Algorithm 1 The Report Noisy Min Algorithm

Require: data $\mathcal{S}$, p-values $p_1, \dots, p_n$ each with sensitivity at most Δ, privacy parameter μ, scale parameter b.
for $j = 1$ to n **do**
  set $\tilde{f}_j = G[G^{-1}(p_j(\mathcal{S})) + Z_j]$, where $Z_j \sim \mathrm{Lap}(0, b\Delta)$;
end for
Output: $j^* = \arg\min_j \tilde{f}_j$ and $\tilde{p}_{j^*} := G[G^{-1}(p_{j^*}(\mathcal{S})) + Z]$, where $Z \sim \mathrm{Lap}(0, b\Delta)$.

The following lemma establishes that the Report Noisy Min Algorithm satisfies μ-GDP, providing the first step toward the overall privacy guarantee.

Lemma 2 *The Report Noisy Min Algorithm, as detailed in Algorithm 1, is μ-GDP for $b = 1/\log\{\Phi(\mu/(2\sqrt{2}))/\Phi(-\mu/(2\sqrt{2}))\}$.*

Algorithm 2 The Mirror Peeling Algorithm

Require: p-values $p_1 \dots, p_n$ each with sensitivity at most Δ, privacy parameter μ, size m, scale parameter b.
let $\mathcal{S} = \{1, \dots, n\}$ be the index set of p-values;
for $j = 1$ to m **do**
  let i_j be the returned index of the report noisy min algorithm applied to partially masked p-values

$$\{\min(p_i, 1 - p_i)\}_{i \in \mathcal{S}},$$

where $Z_j \sim \mathrm{Lap}(0, b\Delta)$;
  let $\tilde{p}_{i_j}$ be the noisy p_{i_j} defined in Algorithm 1 where $Z \sim \mathrm{Lap}(0, b\Delta)$;
  Update $\mathcal{S} = \mathcal{S} \backslash \{i_j\}$;
end for
Output: $\Big\{(i_1, \tilde{p}_{i_1}), \dots, (i_m, \tilde{p}_{i_m})\Big\}$.

For the scale choice $b = \sqrt{10m\log(1/\delta)}/\varepsilon$, Theorem 10 of Xia and Cai (2023) proves an (ε, δ)-DP guarantee for the peeling procedure obtained by composing the Report Noisy Min subroutine over m steps. The following lemma provides the corresponding μ-GDP calibration: with an appropriate alternative choice of the scale parameter b, the same peeling procedure satisfies μ-GDP.

Lemma 3 *The Mirror Peeling Algorithm, as presented in Algorithm 2, is μ-GDP for $b = 1/\log\{\Phi(\mu/(2\sqrt{2m}))/\Phi(-\mu/(2\sqrt{2m}))\}$.*

Proof We first consider the Report Noisy Min Algorithm. Let

$$\varepsilon_0 = \log\left\{\frac{\Phi(\mu/(2\sqrt{2}))}{\Phi(-\mu/(2\sqrt{2}))}\right\}.$$

The report noisy min algorithm sets $b = 1/\varepsilon_0$, and the Laplace noise used in the selection step has scale $\lambda = b\Delta$.

We first prove that the selection step is ε_0-DP. Let

$$M_1(\mathcal{S}) = \arg\min_{j\in[n]}\{f_j(\mathcal{S}) + Z_j\},$$

where $Z_1, \ldots, Z_n \overset{iid}{\sim} \mathrm{Lap}(0, \lambda)$ and we use the notation $f_j(\mathcal{S}) := G^{-1}(p_j(\mathcal{S}))$. Since G is monotone increasing, selecting the minimum of $G(f_j(\mathcal{S}) + Z_j)$ is equivalent to selecting the minimum of $f_j(\mathcal{S}) + Z_j$. By Lemma 2.4 in Dwork et al. (2021), the selection step M_1 is ε_0-DP. Next, we convert this pure-DP guarantee into GDP. By Lemma 4, we know every ε-DP mechanism is

$$2\Phi^{-1}\left(\frac{e^{\varepsilon}}{1+e^{\varepsilon}}\right)\text{-GDP}.$$

Applying this with $\varepsilon = \varepsilon_0$, and using $e^{\varepsilon_0} = \Phi(\mu/(2\sqrt{2}))/\Phi(-\mu/(2\sqrt{2}))$, we get

$$\frac{e^{\varepsilon_0}}{1+e^{\varepsilon_0}} = \frac{\Phi(\mu/(2\sqrt{2}))}{\Phi(\mu/(2\sqrt{2})) + \Phi(-\mu/(2\sqrt{2}))} = \Phi(\mu/(2\sqrt{2})).$$

Therefore,

$$2\Phi^{-1}\left(\frac{e^{\varepsilon_0}}{1+e^{\varepsilon_0}}\right) = 2\Phi^{-1}(\Phi(\mu/(2\sqrt{2}))) = \mu/\sqrt{2}.$$

Thus, the selection step is $\mu/\sqrt{2}$-GDP.

Now consider the second step of the algorithm. After j^* is selected, the algorithm releases

$$\widetilde{p}_{j^*} = G(f_{j^*}(\mathcal{S}) + Z),$$

where $Z \sim \mathrm{Lap}(0, b\Delta)$. By the Laplace mechanism (Theorem 3.6 in Dwork et al. (2014)), it is ε_0-DP and thus $\mu/\sqrt{2}$-GDP by the previous proof. By the composition theorem for GDP (Corollary 2 in Dong et al. (2022)), the joint release $\left\{(i_1, \tilde{p}_{i_1}), \ldots, (i_m, \tilde{p}_{i_m})\right\}$ is μ-GDP. Then, we complete the proof for Lemma 2.

For Algorithm 2, by Lemma 2. For each $j \in \{1, \ldots, m\}$, the joint release $\left(i_j, \widetilde{p}_{i_j}\right)$ is $\mu/\sqrt{m}$-GDP. By the composition theorem for GDP (Corollary 2 in Dong et al. (2022)), the joint release $(j^*, \widetilde{p}_{j^*})$ is μ-GDP. Then, we complete the proof for Lemma 3. ■

The proof of Lemma 3 relies on the following lemma, which is crucial for connecting the (ε, δ)-DP and μ-GDP frameworks.

Lemma 4 *Every ε-DP mechanism is μ-GDP with*

$$\mu = 2\Phi^{-1}\left(\frac{e^{\varepsilon}}{1+e^{\varepsilon}}\right).$$

Proof Let M be an ε-DP mechanism. By (4) in Dong et al. (2022), we know that ε-DP implies f_ε-DP with

$$f_\varepsilon(\alpha) = \max\left\{0, 1 - e^{\varepsilon}\alpha, e^{-\varepsilon}(1-\alpha)\right\}.$$

To show M satisfies μ-GDP, it remains to show that $f_\varepsilon \geq G_\mu(\alpha)$ for all $\alpha \in [0,1]$, where

$$G_\mu(\alpha) = \Phi\left(\Phi^{-1}(1-\alpha) - \mu\right)$$

is the Gaussian tradeoff function. For $\alpha_* = 1/(1+e^\varepsilon)$, the two nonzero linear pieces of f_ε meet at α_*. More precisely,

$$f_\varepsilon = \begin{cases} 1 - e^\varepsilon \alpha, & 0 \leq \alpha \leq \alpha_*, \\ e^{-\varepsilon}(1-\alpha), & \alpha_* \leq \alpha \leq 1. \end{cases}$$

Then, we evaluate $G_\mu(\cdot)$ at three points. By calculation, we have

$$G_\mu(0) = 1, \qquad G_\mu(1) = 0, \qquad G_\mu(\alpha_*) = \alpha_*$$

Because G_μ is convex, its graph lies below the chord joining any two points on its graph. Thus, $G_\mu(\alpha) \leq f_\varepsilon(\alpha)$ for all $\alpha \in [0,1]$. Thus, by the definition of μ-GDP (Definition 4 in Dong et al. (2022)), M is μ-GDP with $\mu = 2\Phi^{-1}\left(\frac{e^\varepsilon}{1+e^\varepsilon}\right)$. ■

2.1 Theorem 9 of Xia and Cai (2023)

In this section, we state the corrected version of Theorem 9 of Xia and Cai (2023). The correction is that the DP-AdaPT algorithm must use Laplace noise for the selection step in the Report Noisy Min Algorithm.

Theorem 5 (Corrected Theorem 9 of Xia and Cai (2023)) *Let $\mu > 0$ be the target privacy parameter and m be the peeling size. Consider the DP-AdaPT algorithm described in Algorithm 4 of Xia and Cai (2023), with the Mirror Peeling step implemented using the Laplace-noise version in Algorithm 2, where the noise variables Z and Z_j are drawn independently from* $\mathrm{Lap}(0, b\Delta)$. *Then the corrected DP-AdaPT algorithm is μ-GDP for $b = 1/\log\{\Phi(\mu/(2\sqrt{2m}))/\Phi(-\mu/(2\sqrt{2m}))\}$.*

Proof By Lemma 3, the corrected Mirror Peeling step is μ-GDP. By the post-processing property of GDP (Proposition 4 in Dong et al. (2022)), the corrected DP-AdaPT algorithm is μ-GDP. ■

The correction above concerns only the privacy analysis of μ-GDP. The finite-sample FDR control argument remains unchanged, since the adaptive thresholding and stopping rules are applied to the same form of released noisy p-values.

References

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